

Report About Pressure Relief Valves According to API 520 Standard – Unit 2

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1. Introduction

Pressure relief valves (PRVs) are essential safety devices used to protect pressure vessels, piping, and equipment from overpressure conditions.

This project aims to verify the compliance of existing pressure relief valves in the plant with the requirements of the **American Petroleum Institute (API) Standard 520 – Sizing, Selection, and Installation of Pressure-Relieving Devices**.

The inspection focuses on:

- Correct valve selection and sizing (API 520 Part I)
- Proper installation and maintenance practices (API 520 Part II)

2. Objectives

1. To evaluate each installed pressure relief valve for compliance with API 520.
2. To identify any deviations that could affect safety or performance.
3. To recommend corrective actions where necessary.
4. To establish a standard inspection checklist for future audits.

3. Scope

This project covers:

- Pressure relief and safety relief valves installed on process vessels, heat exchangers, and piping systems within the unit 2 turbine section.
- Verification of documentation, physical inspection, and comparison with API 520 requirements.

It does **not** include testing or recalibration of valves, which is handled by certified workshops.

4. Applicable Standards and References

- **API Recommended Practice 520 Part I (2000)** – *Sizing and Selection*
- **API Standard 520 Part II (2015)** – *Installation*
- Manufacturer datasheets and test certificates for each PRV

5. Methodology

5.1 Data Collection

For each valve:

- Record tag number, equipment served, fluid type, set pressure, diameter size.
- Review design data: vessel MAWP, design pressure, operating pressure, and temperature.
- Obtain nameplate photographs and test certificates.

5.2 Field Inspection

- Conduct on-site visual inspection following the **API 520 compliance checklist**.
- Observe valve location, inlet and discharge piping configuration, support, venting, and accessibility.

5.3 Evaluation Criteria

Each PRV was checked against key API requirements:

Parameter	Acceptance Criteria (per API 520)
Set Pressure	$\leq$ Vessel MAWP
Inlet Pressure Loss	$\leq$ 3% of set pressure
Back Pressure	$\leq$ 10% (conventional), $\leq$ 30% (balanced)
Valve Type	Suitable for service (gas, vapor, liquid, or two-phase)
Installation	Close to protected equipment, free-draining piping
Isolation Valves	Lock-open, full-bore, properly tagged
Venting / Draining	Adequate and directed away from personnel

5.4 Documentation

All findings were recorded in the inspection checklist and summarized in a compliance matrix.

Serial No.	Description	Valve Identification details	Location / Main Equipment	Nominal Diameter (mm)	Set Pressure (MPa)	Medium	Discharge method	Total Nos. of Safety valves	valve Outlet pressure Mpa	Design Pressure of vessle / Mpa	Pressure loss <3%	back pressure 10%
1	BFP booster pump inlet line safety valves	A 41H -25	0 m level / 6.3m level	25	1.31	Water	atmosphere	3	0.1	1.2	8.4	7.63
6	LP heater 8/7 vent line safety valve	A 41H- 40	6.3 m level / LP heaters 8/7	25	3.95	water	atmosphere	1	0.1	4	1.27	2.53
7	LP heater 6 vent line safety valve	A 41H -40	6.3 m level / LP heaters 6	25	3.95	water	atmosphere	1	0.1	4	1.27	2.53
8	HP heater 3 safety valve	A 48Y-4	6.3 m level / HP heater 3	100	2.09	steam	atmosphere	1	0.1	2.2	5.26	4.78
9	LP heater 6 safety valve	A47H-16C	6.3 m level / LP heater6	100	0.59	steam	atmosphere	1	0.1	0.6	1.69	16.95
10	Condensate extraction pump inlet safety valves	A 42 Y-16C	0 m level / Condensate Extraction system	25	0.3675	Water	atmosphere	2	0.1	0.15	59.18	27.21
12	CCCW heat Exchanger(Demin side)	A 47H-16C	0 m level / CCCW Heat Exchangers	100	1.3	Demin water	atmosphere	2	0.1	1.25	3.85	7.69
13	CCCW heat Exchange (sea water side)	A 47 H-16P	0 m level / CCCW Heat Exchangers	100	0.7	Sea water	atmosphere	2	0.1	1.25	78.57	14.29
14	LP Heater 5 safety valve	A 47 H -16C	12.6 m level / LP heater 05	100	0.5	Steam	atmosphere	1	0.1	0.6	20	20.00
15	LPH 5 Vent safety valve	A 41 H - 40	12.6 m level / LP heater 05	25	3.95	Water	atmosphere	1	0.1	4	1.27	2.53
16	HP Heater 2 safety valve	A 48 Y - 3	12.6 m level / HP heater 02	100	4.56	Steam	atmosphere	1	0.1	4.8	5.26	2.19
17	HP Heater 1 safety valve	A 48 Y -10	12.6 m level / HP heater 01	100	7.315	Steam	atmosphere	1	0.1	7.7	5.26	1.37
18	Auxiliary Header safety valve	A 48 Y -25	16.6 m level / Auxiliary header	250/300	1.68	Steam	atmosphere	1	0.1	1.7	1.19	5.95
19	Deaerator Tank and Deaerator	A 48 Y -25	18.9 m level / Deaerator	150/200	1.17	Steam	atmosphere	4	0.1	1.2	2.56	8.55
20	HP heater 03 Feed water inlet safety valve		Upper area CCCW pumps	20	26.13	Steam	atmosphere	1	0.1	27.5	5.24	0.38

5.5 Results

Safety Valve Name	Complied or Not	Remark	Recommendation
BFP Booster Pump Inlet valve	No	Set P > MWAP Inlet P loss > 3% Not close to protected equipment	Decrease the Set P below 1.2MPa
LP Heater 8/7 Vent line safety valve	Yes	Fully Complied	-
LP Heater 6 Safety valve	Yes	Fully complied	-
HP Heater 3 safety valve	No	Inlet p loss > 3%	Increase the Set P
LP Heater 6 Safety Valve	No	Back P > 10%	Decrease the outlet P
Condensate extraction pump inlet safety valve	No	Set P > MWAP Inlet p Loss > 3% Back P > 10%	Decrease the Set P below 0.2MPa
CCCW heat Exchanger (Demin side)	No	Set P > MWAP	Decrease the Set P below 1.25Mpa
CCCW heat Exchanger (Sea water)	No	Inlet P loss > 3% Back P > 10%	Increase the set P near to 1.25MPa Better to replace with new one
LP Heater 5 safety valve	No	Inlet P loss > 3% Back P > 10%	Increase the set P near to 0.6MPa
LPH 5 vent line safety valve	yes	Fully complied	-
HP Heater 2 Safety valve	No	Inlet P loss > 3%	Increase the set p near to 4.7Mpa
HP Heater 1 safety valve	No	Inlet P Loss > 3%	Increase the set p 7.6Mpa
Auxiliary header safety valve	Yes	Fully Complied	-
Deaerator tank and Deaerator	Yes	Fully complied	-
HP Heater 3 Feed water inlet safety valve	No	Inlet P Loss > 3% Not close to protected equipment	Increase the Set P near to 27MPa

6. Discussion of Results

The evaluation, summarized in the results table, indicate that a significant number of pressure relief valves within the plant are not fully compliant with API 520 standards. The deviations found present potential risks to equipment integrity and personnel safety. The non-compliant findings can be categorized into several key areas:

6.1 Excessive Inlet Pressure Loss

This was the most common finding, affecting numerous critical valves, including those on HP Heaters 1, 2, and 3, the BFP Booster Pump, Condensate extraction pump, and others. The API 520 standard requires inlet pressure loss to be $\leq 3\%$ of the set pressure.

High inlet pressure loss can cause the valve to "chatter" (open and close rapidly), which drastically reduces its flow capacity and can lead to mechanical failure of the valve itself. This condition prevents the valve from properly protecting the equipment. This issue is often linked to improper installation, such as excessive pipe length between the vessel and the PRV, or undersized inlet piping. The findings for the BFP Booster Pump and HP Heater 3 Feed water inlet valve, which were "Not close to protected equipment", support this correlation.

6.2 Set Pressure Exceeding MAWP

A critical safety failure was observed in several valves, including the BFP Booster Pump inlet, Condensate extraction pump, and the CCCW heat exchanger (Demin side), where the valve's set pressure (Set P) was found to be higher than the vessel's Maximum Allowable Working Pressure (MAWP). The acceptance criterion strictly states that the Set Pressure must be $\leq$ Vessel MAWP.

For example, the BFP booster pump inlet valve (Serial 1) has a set pressure of 1.31 MPa, which is greater than the vessel's design pressure (MAWP) of 1.2 MPa. This configuration means the equipment is allowed to be over pressured *before* its safety device activates, defeating the primary purpose of the PRV.

6.3 Excessive Back Pressure

For conventional PRVs discharging to the atmosphere, the back pressure should not exceed 10% of the set pressure. This limit was exceeded in several cases, including the LP Heater 6 safety valve, Condensate extraction pump, CCCW heat exchanger (Sea water), and LP Heater 5 safety valve.

Excessive back pressure on a conventional valve will reduce the effective lift of the valve disc, thereby decreasing its relieving capacity. In a major overpressure event, the valve may not be able to vent fluid fast enough to prevent the system pressure from rising to a dangerous level.

6.4 Fully Compliant Valves

It is noted that several valves were found to be fully compliant with the API 520 standard, including the LP Heater 8/7 vent line, LP Heater 6 vent line, LPH 5 vent line, Auxiliary header, and Deaerator tank valves. These installations can serve as models for corrective actions on other systems.

7. Conclusion

This project successfully evaluated the installed pressure relief valves for compliance with API 520, as outlined in the objectives. The methodology of data collection, field inspection ¹, and evaluation against API criteria was effective.

The primary conclusion is that a significant portion of the unit 2 turbine section pressure relief valves are non-compliant and pose a risk to system integrity. Key deviations were identified in three major categories:

1. Widespread excessive inlet pressure loss, often exceeding 3%.
2. Critically high set pressures, in some cases exceeding the MAWP of the protected equipment.
3. Excessive back pressure on conventional valves, reducing their capacity.

These findings indicate systemic issues in either the initial design, installation, or management of changes for these PRV systems. The recommendations provided are necessary to correct these safety deviations. A standard inspection checklist was also established as a result of this audit, fulfilling all project objectives.

8. Recommendations

Based on the findings and the project's objective to recommend corrective actions, the following measures are proposed:

8.1 Immediate Corrective Actions (High-Risk)

- Set Pressure Adjustment: Valves identified with "Set P > MWAP" must be corrected immediately.
 - BFP Booster Pump Inlet: Adjust set pressure to be ≤ 1.2 MPa.
 - Condensate Extraction Pump: Adjust set pressure to be ≤ 0.15 MPa.
 - CCCW Heat Exchanger (Demin): Adjust set pressure to be ≤ 1.25 MPa.
- These adjustments must be performed and certified by authorized workshops, as testing is outside the scope of this project.

8.2 Engineering Review and Modification

- Inlet Piping: All valves that failed the $\leq 3\%$ inlet pressure loss criterion must undergo a detailed hydraulic engineering review. This applies to HP Heaters 1, 2, and 3, LP Heater 5, Condensate pump, CCCW (Sea water), and others. Modifications will likely be required, such as relocating valves closer to the equipment (as noted for the BFP Booster and HP Heater 3 Feed water valves) or increasing the diameter of inlet piping per API 520 Part II.
- Outlet Piping: Valves failing the $\leq 10\%$ back pressure criterion (e.g., LP Heater 6, Condensate, CCCW Sea water) require a similar review of their discharge piping.
- Valve Type Re-evaluation:
 - For the CCCW heat exchanger (Sea water), which failed on multiple criteria, replacement with a new, correctly sized valve is recommended.
 - For systems where high back pressure (e.g., > 10) is unavoidable due to discharge manifold design, consideration should be given to replacing the conventional PRVs with balanced-bellows type valves, which can tolerate higher back pressure (up to 30%).

8.3 Future Compliance

- The standard inspection checklist developed during this project should be formally adopted for all future PRV procurement, installation, and routine audits to prevent recurrence of these non-compliance issues.